

Retail Demand Forecasting with Classical Time Series

Models:

Performance and Structural Limitations

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1 Introduction

Retail demand forecasting is a supply chain problem with asymmetric consequences. Order too much and capital is tied up in unsold inventory; order too little and shelves run empty, losing both the immediate sale and potentially the customer. At Walmart’s scale, even a modest improvement in forecast accuracy across thousands of stores and products translates to meaningful cost reductions and service improvements.

The forecasting challenge is harder than it first appears, for two structural reasons. First, observed sales are not the same as true demand. When a shelf empties and a customer cannot find the product, the lost sale is never recorded. Without inventory and stockout logs, a zero in the sales data could mean either that no customer wanted the product, or that the product was unavailable to buy. This distinction matters for modeling: a model trained on observed sales is learning from a censored signal, not from true demand.

Second, retail demand at the individual product and store level is highly intermittent. Most product–store–days record zero sales, and on the days that do record a sale, the quantity is small and erratic. This zero inflation violates the distributional assumptions behind classical Gaussian time series models and makes reliable interval estimation at the individual series level very difficult.

This report addresses the following research question:

How well can classical time series models — Naive, Moving Average, SARIMA, and Prophet — capture retail demand across aggregate, store, and individual SKU levels, and what structural limitations do they reveal?

The analysis is structured around three levels of the demand hierarchy: the full platform aggregate (all products and stores combined), three representative stores (CA_1, CA_3, TX_2), and one individual product–store series. Each level surfaces different model failure modes, and together they define the boundary of what univariate classical forecasting can achieve on this dataset.

2 Data Description

2.1 Dataset Overview

The M5 Forecasting dataset contains actual Walmart retail transactions spanning January 29, 2011 through April 24, 2016 (approximately 5.25 years). Three files provide all the raw material for this project.

`sales_train_validation.csv` is the core file: 30,490 rows in wide format, one column per day (`d_1` through `d_1913`), representing daily unit sales for every product–store combination. Each row is a *series*: a unique pairing of one product with one store, tracked continuously over time. Reshaping from wide to long format produces 58,327,370 observations.

`calendar.csv` maps day identifiers to calendar dates and provides event flags (holiday names, event categories) and binary SNAP distribution indicators for California, Texas, and Wisconsin. `sell_prices.csv` supplies weekly sell prices per product per store. Revenue is computed as $\text{units_sold} \times \text{sell_price}$ and serves as the primary demand proxy throughout; it is not directly observed in the data.

All modeling in this project uses monthly aggregations of the daily data. Monthly aggregation aligns with the 12-month seasonal period appropriate for SARIMA and Prophet, and substantially reduces zero inflation, making the Gaussian assumptions behind these models more defensible than they would be on daily data.

Table 1: M5 dataset dimensions.

Dimension	Value
Product-store series	30,490
Unique products	3,049
Stores	10 (CA, TX, WI)
Date range	Jan 29, 2011 – Apr 24, 2016
Training days	1,913
Total long-format rows	58,327,370

2.2 The Stockout Ambiguity

A fundamental limitation of the dataset is the absence of inventory and stockout records. A zero in the observed sales data is ambiguous: it could reflect genuine zero demand (the product was available but no customer wanted it) or a supply-side censoring event (the product was out of stock, so sales were prevented regardless of demand). These two cases are statistically indistinguishable from the sales file alone.

The practical scale of this problem is illustrated by the zero-streak analysis. Across a sample of 2,000 product-store series, the mean longest consecutive zero-sales streak was 430 days. Forty-four percent of series had a streak exceeding 365 days. Runs of this length are implausible as pure intermittent demand and strongly suggest that structural supply gaps (product not stocked, delisted, or not yet introduced at that store) are the dominant source of zeros. All results in this report should be read with this caveat in mind: the models are approximating a censored demand proxy, not recovering true latent demand.

2.3 Zero Inflation in Observed Sales

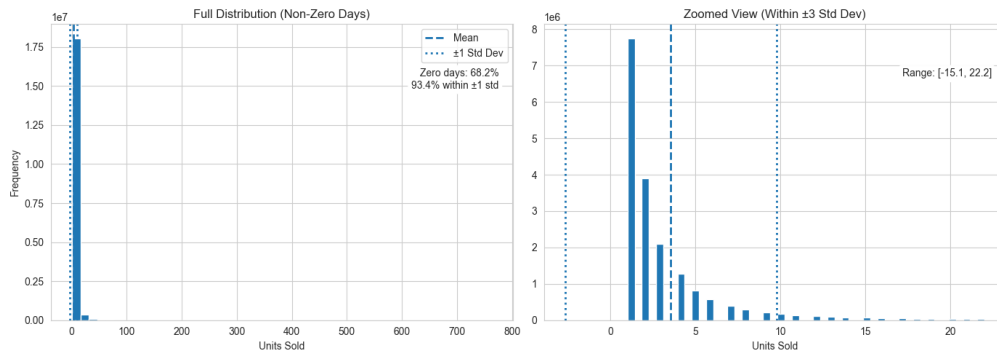


Figure 1: Distribution of observed units sold on non-zero product-store-days (left: full range; right: zoomed to within 3 standard deviations). On non-zero days the median is 2 units and the distribution is heavily right-skewed (skewness 11.89), with a maximum of 763 units recorded at a single store in one day. Aggregation across series is essential to recover a stable demand signal from these sparse observations.

As Figure 1 shows, 68.2% of all product-store-day observations record zero sales. Among non-zero days, the mean is 1.13 units with a standard deviation of 3.87, indicating heavy overdispersion relative to what a Gaussian model assumes. The implication for this project is direct: individual series models will carry wide and unreliable confidence intervals, while aggregate models benefit substantially from the smoothing effect of summing across 30,490 series.

2.4 Series Completeness

Figure 2 shows that only 8.1% of series (2,469 of 30,490) record observed demand in every month of the usable history. The remaining 91.9% contain gaps that conflate supply absence with demand absence. SARIMA requires a continuous time index and is therefore viable only for complete series. The representative individual series used

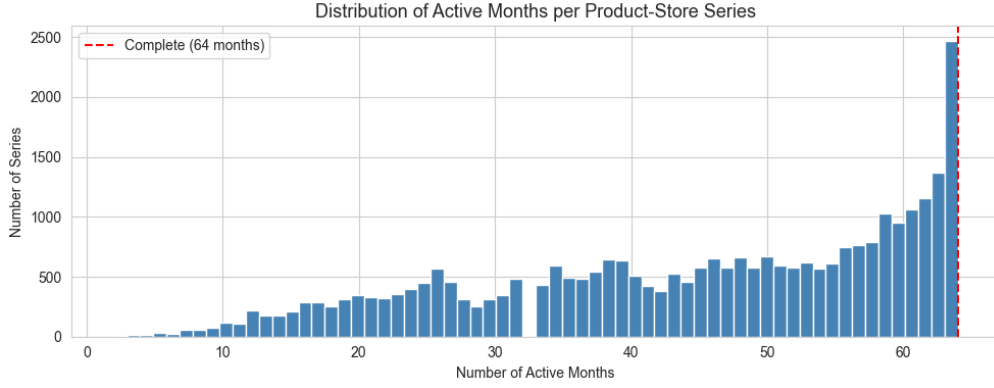


Figure 2: Distribution of active months per product-store series (defined as months with at least one observed sale). Only 2,469 of the 30,490 series (8.1%) show demand in all 64 usable months. The wide spread reflects product lifecycle dynamics: launches, delists, and store-level stocking decisions rather than data quality issues.

throughout this report, FOODS_3_163_CA_3, is selected from these 2,469 complete series as the one closest to the median total observed revenue, averaging approximately \$120 per month.

3 Exploratory Data Analysis

Each finding in this section connects directly to a modeling decision made in later sections. The goal is not to describe the data exhaustively but to identify the structural properties that determine which models are appropriate and where they will fail.

3.1 Revenue Concentration and Demand Density

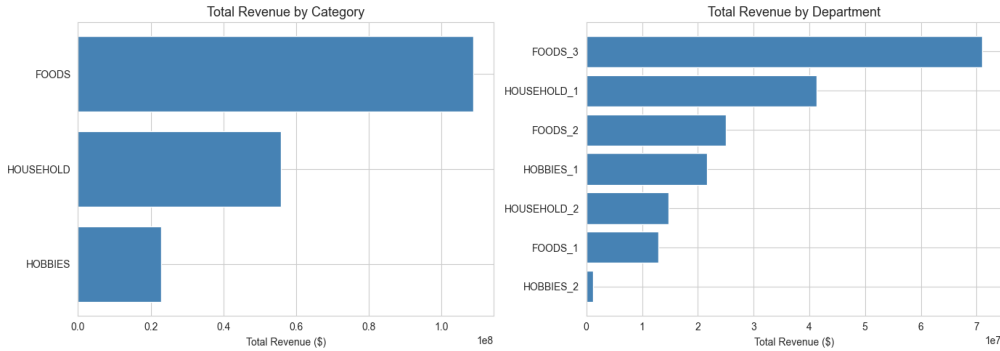


Figure 3: Total observed revenue by product category (left) and department (right) over the full 2011–2016 period. FOODS accounts for 58% of platform revenue; FOODS_3 alone is 37.8%. HOBBIES_2 at 0.63% is too sparse for reliable individual-level modeling.

Figure 3 shows strong revenue concentration. FOODS accounts for 58% of total observed revenue, with FOODS_3 the single largest department at 37.8%. High revenue density in FOODS reflects frequent, high-volume transactions that produce the continuous demand signal statistical models require. HOBBIES_2 at 0.63% is so sparse that individual series models produce unreliable approximations.

Store-level concentration mirrors this pattern. CA_3, the highest-revenue store, accounts for 17.1% of platform revenue; CA_4, in the same state, accounts for only 6.5%. This near-threefold difference between stores in the same state confirms that state-level features are insufficient: store-level encoding is required for any model operating below the platform aggregate.

3.2 Seasonality and Trend in Aggregate Demand

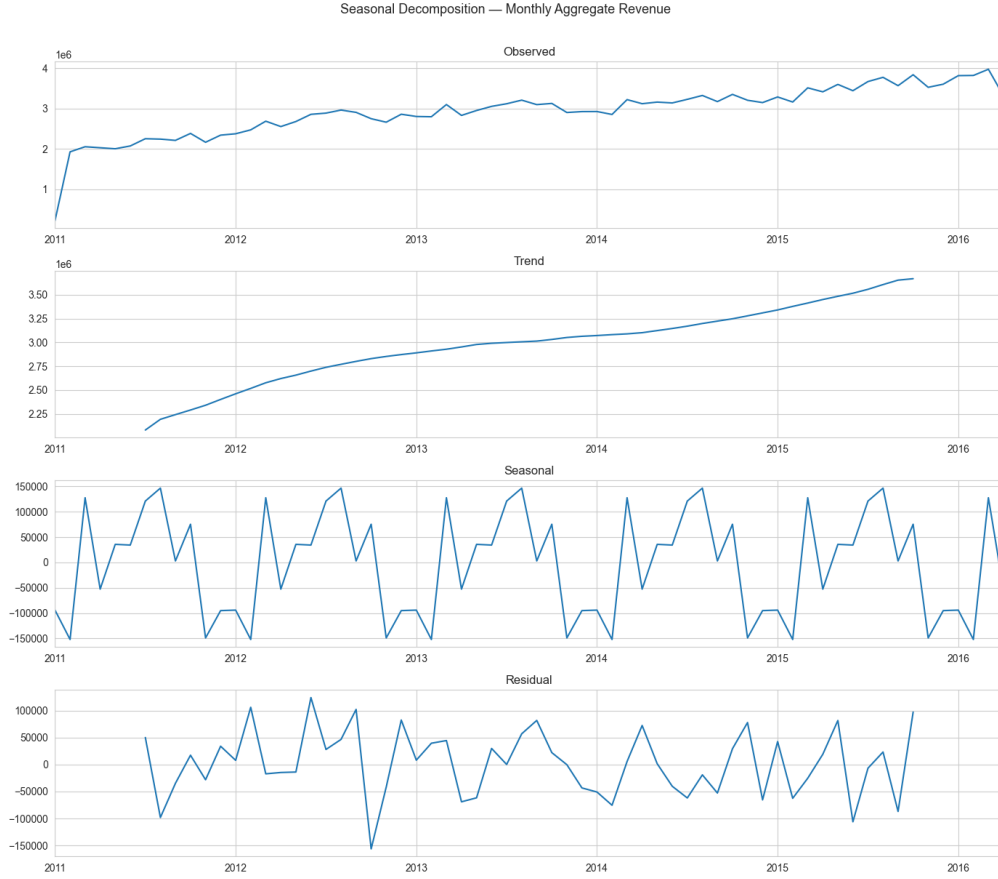


Figure 4: Additive seasonal decomposition of monthly aggregate observed revenue. Trend (second panel) shows steady growth from approximately \$2M/month in 2011 to \$4M/month by 2016. The seasonal component (third panel) is modest relative to trend; August and March carry the strongest positive deviations (+\$147K and +\$128K respectively). The residual (bottom panel) is approximately white noise.

Figure 4 reveals two properties of the aggregate series that directly shape model selection. First, the trend dominates: observed aggregate demand roughly doubled over five years at a sustained rate with no structural reversal. Any model that cannot extrapolate this trend forward will underestimate demand progressively at longer horizons. This is the mechanism behind SARIMA’s long-horizon underperformance, examined in Section 7.

Second, the seasonal amplitude is small relative to trend. The strongest seasonal swing is approximately \$150K against a \$3M monthly baseline (roughly 5%). The pattern is real and statistically recoverable, but models that optimise seasonal fit at the expense of trend extrapolation will still degrade at long forecast horizons.

3.3 SNAP Effects

SNAP (Supplemental Nutrition Assistance Program) benefits are distributed to eligible households on specific calendar days. Recipients concentrate spending immediately after distribution, producing a predictable demand pulse at grocery stores. As Figure 5 shows, the uplift in observed FOODS revenue ranges from +10.3% in California to +32.5% in Wisconsin, with state-specific timing because each state operates its own distribution schedule.

This signal is both strong and foreseeable: the SNAP calendar is known in advance, making it a predictive external demand driver rather than a historical correlation. SARIMA and Prophet read only the observed revenue series and have no mechanism to incorporate the SNAP calendar as an explicit input. This is one of the primary drivers of the

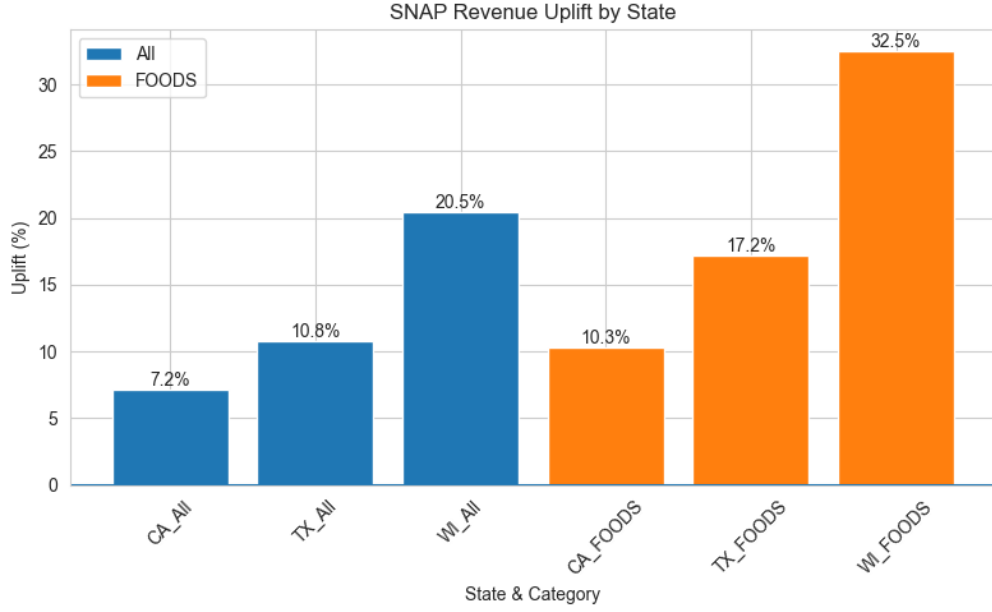


Figure 5: Observed revenue uplift on SNAP benefit distribution days, by state, for all categories and for FOODS only. The SNAP effect is stronger in FOODS because benefits are restricted to food purchases. Wisconsin FOODS shows the largest uplift (+32.5%); California FOODS the smallest (+10.3%).

signal ceiling quantified in Section 10.

3.4 Price Elasticity

Figure 6 illustrates how aggregation conceals price sensitivity. At the department level, correlations between price changes and demand changes are near zero. This is an example of reverse causality: Walmart marks down slow-selling products, so price drops and weak demand co-occur because both are caused by poor underlying demand, not because discounts suppress sales. Aggregating across thousands of products with different baseline demand levels amplifies this distortion until the real signal disappears.

At the individual product–store level, once the same product’s own price changes are tracked over time, the distortion disappears and asymmetric elasticity is visible. A 10% price drop at FOODS_3_383 in CA_3 is associated with approximately 77% more observed units ($r = 0.553$), while price increases show a much weaker response ($r = 0.180$). This asymmetry is consistent with stocking-up behaviour on discounts.

The modeling implication is that price features must be engineered at the product–store level. Department-average price changes carry no predictive value ($r < 0.13$), and signed features are necessary to capture the asymmetric demand response. Like SNAP, this signal is entirely invisible to univariate time series models.

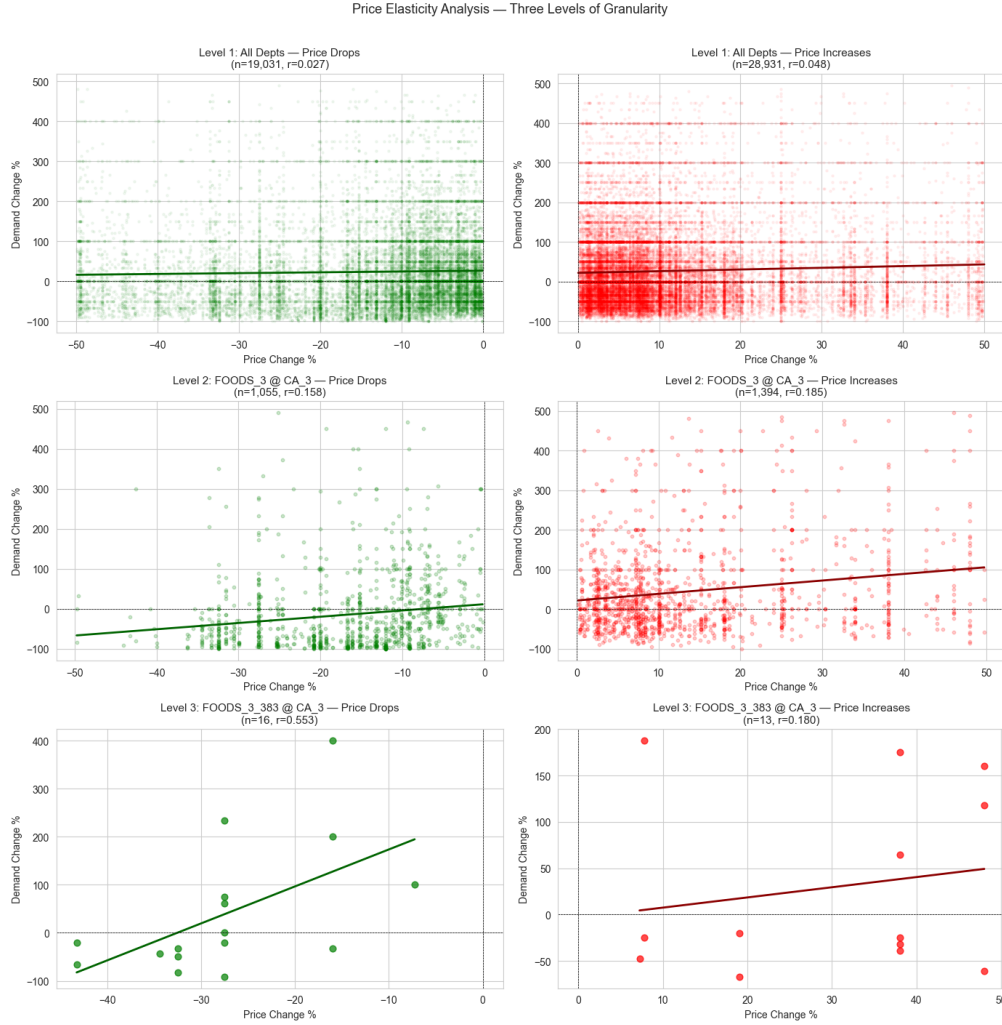


Figure 6: Price elasticity analysis at three granularity levels (department, store-department, individual product-store). Department-level correlations are near zero due to reverse causality: Walmart discounts slow-moving products, so price drops and low demand co-occur without a direct demand relationship. At the individual product-store level the true asymmetric demand response emerges ($r = 0.553$ for price drops, $r = 0.180$ for price increases at FOODS_3_383 / CA_3).

4 Time Series Foundations

4.1 Stationarity

A time series is *stationary* if its mean and variance remain stable over time. A flat, noisy series with no persistent direction is stationary. A series with an upward trend is not: its mean keeps rising, violating the stability assumption. Stationarity matters because SARIMA's parameter estimation assumes the series fluctuates around a fixed level. When that assumption fails, the estimated AR and MA coefficients are unreliable and confidence intervals are poorly calibrated.

The Augmented Dickey-Fuller (ADF) test formalises this check. The null hypothesis is that the series has a unit root (i.e., is non-stationary). A p-value below 0.05 rejects the null, confirming stationarity.

Applied to the two series modeled in this project:

- **Aggregate monthly revenue:** ADF p-value of 0.318 on the raw series, non-stationary. The five-year upward trend violates the stable-mean assumption and must be removed before fitting.

- **Representative series (FOODS_3_163_CA_3):** ADF p-value of 0.0004, already stationary. The individual product–store series fluctuates around a stable mean with no persistent long-run drift.

4.2 Differencing

Differencing transforms a non-stationary series into a stationary one by replacing each observation with the change from the previous period. Two types are relevant here.

First differencing ($d = 1$) subtracts consecutive values, converting demand levels into period-over-period changes. This removes a linear trend.

Seasonal differencing ($D = 1$ at lag m) subtracts the value from the same period one year prior:

$$\nabla_m y_t = y_t - y_{t-m} \quad (1)$$

where $m = 12$ for monthly data with annual seasonality. This removes year-over-year drift that first differencing alone cannot fully address. Applied to the aggregate series, seasonal differencing alone reduces the ADF p-value from 0.318 to approximately 0.000, confirming stationarity without requiring non-seasonal differencing ($d = 0$, $D = 1$).

The key implication is that after seasonal differencing the model operates on year-over-year changes rather than raw levels. This removes the trend from the fitting procedure, which is exactly what is needed, but it also means the model extrapolates by adding predicted changes back onto the last observed level rather than projecting a trend slope forward. At long horizons, this produces mean reversion, a property examined in detail in Section 7.

5 ACF and PACF Analysis

5.1 Interpreting ACF and PACF

The *Autocorrelation Function* (ACF) measures how correlated a time series is with lagged versions of itself. A significant spike at lag k means that the value k periods ago carries predictive information about the current value.

The *Partial Autocorrelation Function* (PACF) isolates the direct relationship at each lag by removing the indirect influence of intermediate lags. A significant PACF spike at lag p that drops sharply to zero thereafter indicates an AR(p) process. A gradual decay in the ACF with a sharp cutoff in the PACF after lag p confirms this AR interpretation.

Conversely, a sharp ACF cutoff after lag q with a gradually decaying PACF indicates an MA(q) process. In practice, ARMA processes produce mixtures of these patterns, and the ACF/PACF plots narrow the parameter search space rather than determining orders definitively. The AIC grid search in Section 7 confirms the final selection.

Significant spikes at lag 12 (or multiples of 12) in monthly data indicate seasonal structure that seasonal AR (P) or seasonal MA (Q) terms are needed to capture.

5.2 Results for the Aggregate Series

Figure 7 (top row) shows the ACF and PACF for the aggregate series after applying seasonal differencing ($D = 1$, lag 12). The PACF shows one significant spike at lag 1 that cuts off sharply, suggesting $p = 1$ as the AR candidate. The ACF decays gradually over lags 1–5 rather than cutting off cleanly, which is consistent with an AR-driven process rather than a pure MA process, suggesting $q = 0$ as the initial candidate. Neither the ACF nor the PACF shows

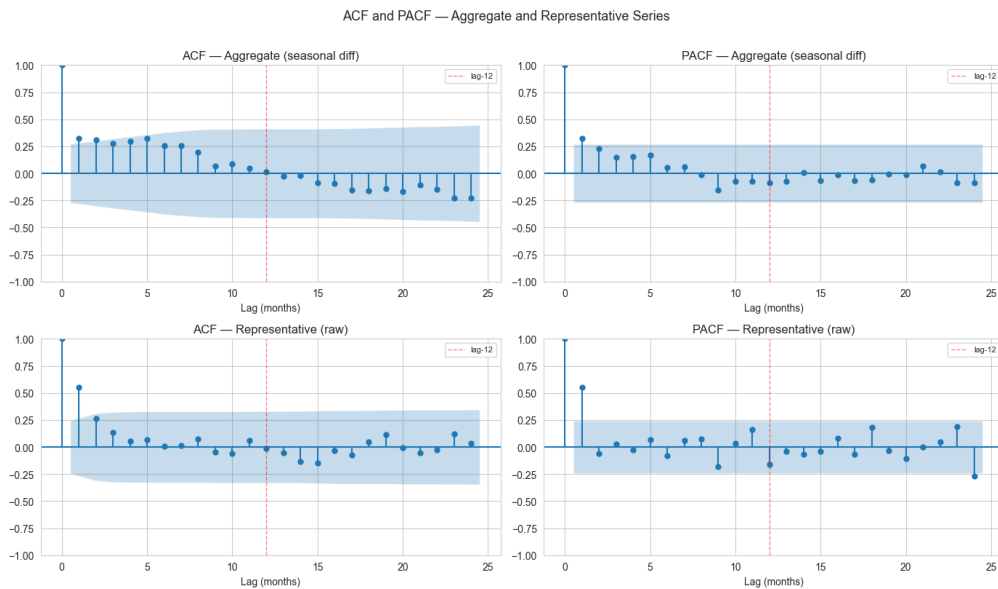


Figure 7: ACF and PACF plots for the aggregate series after seasonal differencing (top row) and the representative individual series on its raw values (bottom row). Shaded bands indicate 95% confidence bounds; bars extending beyond the band are statistically significant.

significant spikes at lag 12, confirming that seasonal differencing has removed the annual structure and that $P = 0$ is appropriate as a starting point.

The AIC grid search subsequently selects $\text{SARIMA}(2, 0, 1)(0, 1, 1)_{12}$, extending the ACF/PACF candidates slightly. The difference between the top two AIC values is less than 0.01, indicating the model is not sensitive to small variations in the AR order around the selected configuration.

5.3 Results for the Representative Series

For the representative individual series (bottom row of Figure 7), the raw series is already stationary ($d = 0$). The PACF shows one significant spike at lag 1 with a sharp cutoff, suggesting $p = 1$. The ACF shows significant values at lags 1–2 before cutting off, suggesting $q = 1$. No significant spikes appear at lag 12 on the undifferenced series.

The AIC grid search selects $\text{SARIMA}(0, 0, 1)(0, 1, 1)_{12}$, dropping the AR term entirely and adding seasonal differencing. The absence of an AR term in the final model is informative: on a noisy, sparse individual series, last month’s observed sales value does not add direct predictive power once the MA error correction term is included. Error correction ($q = 1$) is more useful than momentum ($p = 1$) when the month-to-month signal-to-noise ratio is low. The AIC grid search resolves this ambiguity empirically rather than relying on the ACF/PACF heuristics alone.

6 Baseline Models

Before fitting statistical models, two naive baselines establish the performance floor that every subsequent model must clear to justify its added complexity.

Naive (persistence) forecast: next month’s demand is set equal to the most recently observed value. This is the simplest possible approximation and surprisingly competitive on trending series, where the most recent observation is already the best single-point estimate of the next period.

Simple Moving Average, SMA(3): the forecast is the mean of the three most recent observations. Smoothing is

expected to reduce noise, but on a strongly trending series the three-month average systematically lags the current level, pulling the forecast below where demand actually lands.

Both baselines are evaluated on the held-out 12-month test period (February 2015 through January 2016) using the same metrics applied to all models: Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE). MAPE is the primary comparison metric because it is unit-free and directly comparable across the aggregate, store, and individual series despite their very different revenue scales.

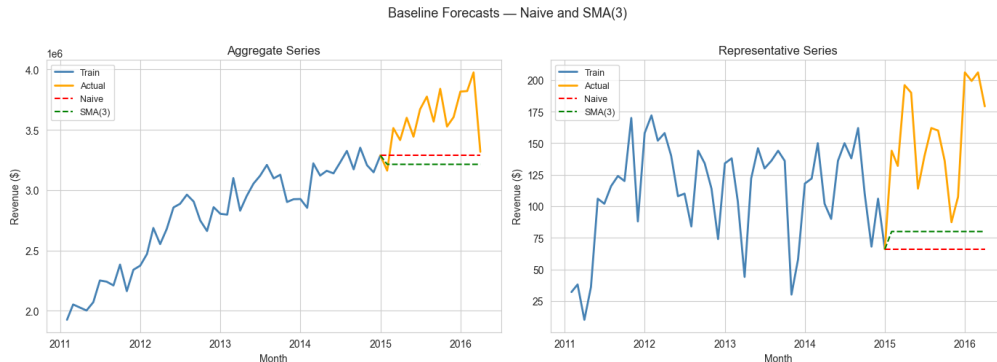


Figure 8: Naive and SMA(3) baseline forecasts against actual observed demand for the aggregate series (left) and the representative individual series (right). On the aggregate series, the Naive forecast (flat red dashed line) is competitive because the upward trend makes the most recent observation the best available estimate. SMA(3) systematically underestimates by averaging in older, lower values.

Figure 8 shows the baseline forecasts against actuals. On the aggregate series, the Naive forecast achieves a MAPE of 8.96% and SMA(3) achieves 10.71%. Naive outperforms the smoothed baseline because the aggregate demand trend is strong enough that the most recent observed value is a better predictor than any backward-looking average. SMA(3) introduces a systematic downward bias by blending in the lower values from two and three months prior.

This result has an important implication: aggregation smooths out series-specific variability across tens of thousands of series, leaving a clean, trending signal that even simple heuristics approximate reasonably well. The baseline MAPEs of 9–11% on the aggregate series are harder to beat than they appear, because the dominant structure (the trend) is already captured by the persistence assumption. SARIMA and Prophet add value primarily through their seasonal correction and, in Prophet’s case, through explicit trend extrapolation.

On the individual representative series the situation is reversed. Naive achieves 55.29% MAPE and SMA(3) achieves 45.80%, both poor because month-to-month observed demand is too erratic for a flat heuristic to track. The baseline is so weak here that any model capturing the annual seasonal pattern will clear it easily. The real question at the individual level is not whether statistical models beat the naive baseline but how large the irreducible gap remains after capturing all available time series structure.

7 SARIMA Model

7.1 Model Formulation

SARIMA (Seasonal AutoRegressive Integrated Moving Average) extends the classical ARIMA framework to handle periodic structure. The full model is written $\text{SARIMA}(p, d, q)(P, D, Q)_m$, where the non-seasonal component captures short-run autocorrelation and the seasonal component captures structure repeating every m periods. For monthly data with annual seasonality, $m = 12$.

The general SARIMA equation combines non-seasonal and seasonal AR/MA polynomials with differencing operators:

$$\Phi_P(B^m) \phi_p(B) (1 - B^m)^D (1 - B)^d y_t = \Theta_Q(B^m) \theta_q(B) \varepsilon_t \quad (2)$$

where B is the backshift operator ($By_t = y_{t-1}$), $\phi_p(B)$ and $\theta_q(B)$ are the non-seasonal AR and MA polynomials of orders p and q , $\Phi_P(B^m)$ and $\Theta_Q(B^m)$ are the seasonal AR and MA polynomials of orders P and Q , and $(1 - B)^d(1 - B^m)^D$ applies the required non-seasonal and seasonal differencing. The AR component uses past observed values as predictors; the MA component uses past forecast errors, making the model self-correcting. The residuals ε_t are assumed to be independently and identically distributed with zero mean.

7.2 Evaluation Metrics

Three metrics are used consistently across all models and series. *RMSE* (Root Mean Squared Error) is the square root of the mean squared deviation between forecast and actual; it penalises large errors more than small ones and is expressed in the same units as the target (dollars of revenue). *MAE* (Mean Absolute Error) is the average absolute dollar error per month, less sensitive to outliers than RMSE. *MAPE* (Mean Absolute Percentage Error) expresses the average error as a percentage of the actual observed value:

$$\text{MAPE} = \frac{1}{n} \sum_{t=1}^n \left| \frac{y_t - \hat{y}_t}{y_t} \right| \times 100 \quad (3)$$

MAPE is the primary metric throughout this report because it is unit-free and directly comparable across the aggregate, store, and individual series despite their very different revenue scales. A MAPE of 7% means the model's forecast is wrong by approximately 7 cents per dollar of actual observed revenue, on average across the test period.

7.3 Parameter Selection via AIC

Model order is selected by minimising the Akaike Information Criterion on the training data only:

$$\text{AIC} = -2 \log(\hat{L}) + 2k \quad (4)$$

where $\hat{L}$ is the maximised likelihood and k is the number of free parameters. AIC rewards goodness of fit while penalising complexity. It is the appropriate selection criterion here because cross-validation on only 48 training months would consume too large a fraction of the available history. The test set is never used during parameter selection. The grid search explores $p \in \{0, 1, 2\}$, $d \in \{0, 1\}$, $P \in \{0, 1\}$, $D \in \{0, 1\}$, $Q \in \{0, 1\}$ for the aggregate series. The selected orders are summarised in Table 2.

Table 2: SARIMA orders selected by AIC grid search on 48-month training data.

Series	Order (p, d, q)	Seasonal $(P, D, Q)_{12}$	Key decision
Aggregate	(2, 0, 1)	(0, 1, 1) ₁₂	$d=0$, $D=1$ from ADF; AIC extended AR to $p=2$
Representative	(0, 0, 1)	(0, 1, 1) ₁₂	AR term dropped; error correction dominates on sparse series

7.4 Aggregate Series Results

Figure 9 shows the aggregate series forecast. Table 3 reports the test-period metrics against the naive baseline established in Section 6.

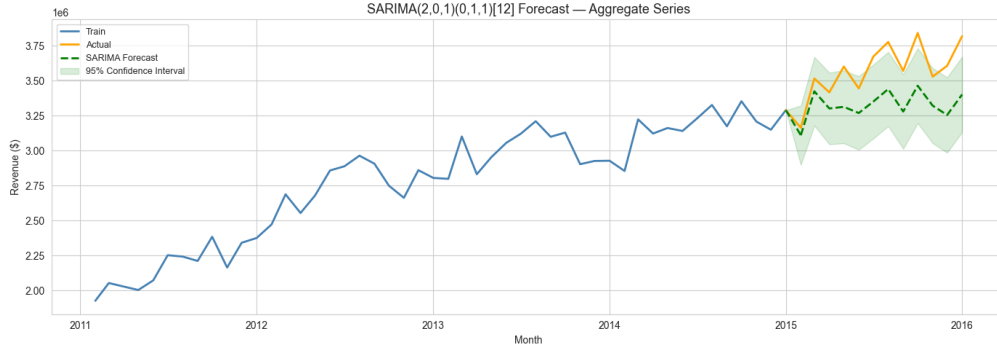


Figure 9: $SARIMA(2,0,1)(0,1,1)_{12}$ forecast against actual observed aggregate revenue over the 12-month test period (Feb 2015 – Jan 2016). The seasonal shape is correctly captured in early months; approximation error grows progressively at longer horizons as mean reversion pulls predictions below the actual upward trajectory.

Table 3: SARIMA aggregate series performance vs. Naive baseline (12-month test period).

Model	RMSE	MAE	MAPE
Naive	\$380,051	\$332,241	8.96%
SMA(3)	\$443,310	\$396,285	10.71%
$SARIMA(2,0,1)(0,1,1)_{12}$	\$277,220	\$252,147	6.91%

SARIMA achieves a MAPE of 6.91%, reducing the average monthly forecast error to approximately 7 cents per dollar of observed revenue. Compared to the Naive baseline, this is a 2.05 percentage point improvement and a \$102,831 reduction in average monthly dollar error (RMSE drops from \$380,051 to \$277,220). The model justifies its added complexity: capturing the annual seasonal structure and short-run autocorrelation produces a meaningfully better approximation than a flat heuristic.

The error pattern over the 12-month horizon, however, reveals the structural limitation introduced in Section 4. Errors in February through April 2015 are 1.7–3.4%, reflecting accurate approximation close to the training window. By January 2016, the error reaches 10.9%. This progressive degradation is caused by mean reversion: after seasonal differencing removes the trend from the fitting procedure, multi-step forecasts are pulled back toward the training-period mean rather than continuing the observed upward trajectory. This is an inherent property of stationary AR models, not a parameter tuning issue. Increasing the AR order or adjusting the MA term does not resolve it because the root cause is the stationarity transformation itself.

7.5 Representative Series Results

Table 4: SARIMA representative series performance vs. baselines (12-month test period).

Model	RMSE	MAE	MAPE
Naive	\$98.48	\$91.31	55.29%
SMA(3)	\$85.66	\$77.31	45.80%
$SARIMA(0,0,1)(0,1,1)_{12}$	\$54.41	\$37.39	22.22%

Figure 10 and Table 4 show the individual series results. SARIMA reduces MAPE from 45.80% (SMA(3) baseline) to 22.22%, a 51% relative improvement and a \$31.25 reduction in average monthly dollar error. At 22 cents of error per dollar of actual observed revenue on average, this is a meaningful improvement for an individual product series with a median of 2 units sold per day.

Most months fall in the 1–20% error range. Three stand out as large misses: April 2015 (54.9%), May 2015 (47.3%),

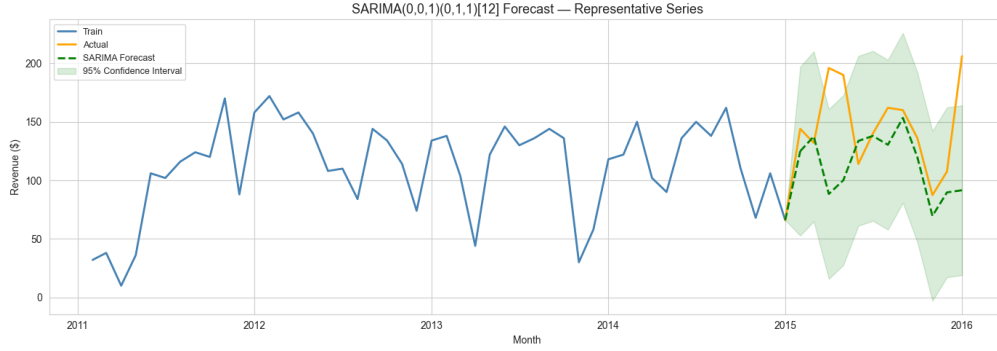


Figure 10: $SARIMA(0,0,1)(0,1,1)_{12}$ forecast for the representative individual series ($FOODS_3_163_CA_3$). The model captures the annual seasonal shape but produces errors exceeding 40% in three months where external demand drivers (price changes, SNAP timing) dominated observed sales.

and January 2016 (55.6%). These spikes indicate months where observed demand was driven by external factors, such as a price promotion or a shift in SNAP distribution timing, that are not present in the revenue time series the model reads. These three months are examined more closely in Section 10.

8 Prophet Model

8.1 Model Formulation

Prophet (Taylor and Letham, 2018) decomposes a time series into three additive components:

$$y(t) = g(t) + s(t) + h(t) + \varepsilon_t \quad (5)$$

where $g(t)$ is the trend, $s(t)$ captures seasonality, $h(t)$ encodes holiday effects, and ε_t absorbs residual noise.

Trend. $g(t)$ is modelled as a piecewise linear function. Prophet automatically identifies changepoints in the training history where the trend slope shifts, fits a separate segment after each, and extrapolates the final segment’s slope into the forecast horizon. This is the critical structural difference from SARIMA: rather than removing trend through differencing and then mean-reverting, Prophet explicitly models the trend and projects it forward. The *changepoint prior scale* (CPS) hyperparameter controls flexibility.

Seasonality. The seasonal component is a Fourier series:

$$s(t) = \sum_{n=1}^N \left[a_n \cos\left(\frac{2\pi nt}{P}\right) + b_n \sin\left(\frac{2\pi nt}{P}\right) \right] \quad (6)$$

where P is the seasonal period and the coefficients a_n , b_n are estimated from training data. This requires no ACF/PACF analysis or lag-order selection.

Holiday effects. The $h(t)$ component accepts a user-supplied dataframe of named events with date windows. Prophet estimates a separate effect for each event, directly encoding the holiday-driven demand shifts quantified in the EDA.

Seasonality mode. In *additive* mode the seasonal swing is a fixed dollar amount regardless of trend level. In *multiplicative* mode the swing is a percentage of the trend, growing as the demand baseline grows. For the aggregate series, cross-validation selected multiplicative: as observed revenue doubled over five years, the seasonal amplitude

grew proportionally. For individual and store series, additive was selected, reflecting more stable demand baselines at those levels.

8.2 Hyperparameter Selection

Hyperparameters are selected via Prophet’s built-in cross-validation on training data only. A sliding window advances the cutoff, evaluates on a 12-month horizon, and selects the configuration minimising RMSE. The test set is not touched during this process. Table 5 shows the final configurations.

Table 5: Prophet hyperparameters selected by cross-validation on 48-month training data.

Series	CPS	Mode	Key finding
Aggregate	0.5	Multiplicative	Seasonal amplitude grows with trend level
Representative	0.01	Additive	Stiff trend avoids overfitting to noise spikes
CA_1	0.01	Multiplicative	Smooth trajectory; low flexibility optimal
CA_3	0.01	Multiplicative	Same structure as CA_1
TX_2	0.05	Additive	Higher CV error signals dominant external drivers

8.3 Aggregate Series Results

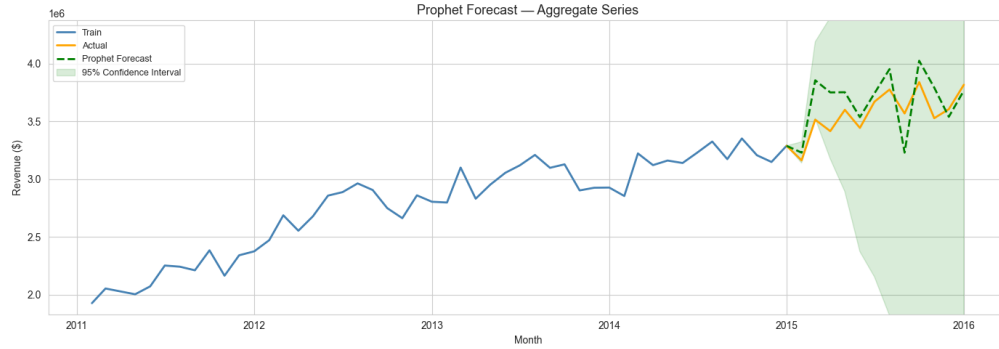


Figure 11: Prophet forecast ($CPS = 0.5$, multiplicative seasonality) against actual observed aggregate revenue over the 12-month test period. Prophet’s piecewise linear trend extrapolates the upward trajectory, keeping long-horizon errors small where SARIMA degrades. The 95% confidence interval widens progressively because uncertainty is scaled as a percentage of the trend level in multiplicative mode, growing alongside the demand baseline.

Table 6: Prophet aggregate series performance vs. all baselines (12-month test period).

Model	RMSE	MAE	MAPE
Naive	\$380,051	\$332,241	8.96%
SMA(3)	\$443,310	\$396,285	10.71%
SARIMA(2, 0, 1)(0, 1, 1) ₁₂	\$277,220	\$252,147	6.91%
Prophet (CPS=0.5, mult.)	\$209,726	\$178,567	5.02%

Figures 11 and 12 and Table 6 show the aggregate results. Prophet achieves a MAPE of 5.02%, beating SARIMA by 1.89 percentage points and the Naive baseline by 3.94 percentage points. In dollar terms, average monthly error falls to \$178,567 (MAE), a \$73,580 reduction compared to SARIMA and a \$153,674 reduction compared to Naive. For every dollar of observed aggregate revenue in the test period, Prophet’s forecast is off by approximately 5 cents on average.

The source of the improvement over SARIMA is directly traceable. Where SARIMA’s error grows to 10.9% by January 2016, Prophet holds to 1.3% on the same month because its piecewise linear trend continues the slope

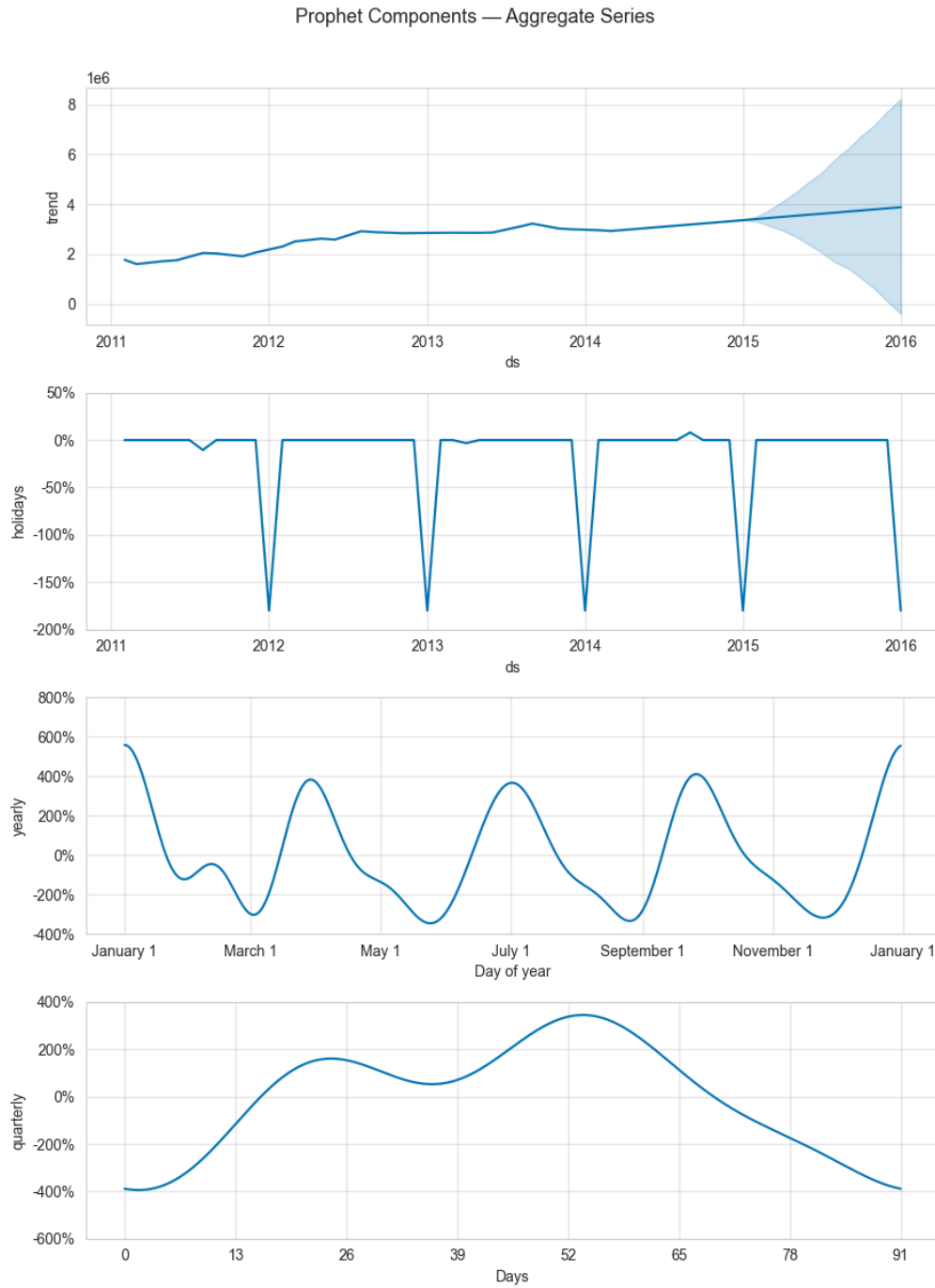


Figure 12: Prophet component decomposition for the aggregate series. The trend panel confirms steady, sustained demand growth over the training period. The yearly seasonality panel recovers the August and March peaks identified in the EDA. The holiday panel encodes event-driven demand shifts including the SuperBowl and Labor Day uplifts quantified in Section 3.

learned from training rather than reverting toward the historical mean. The trade-off is at short horizons: March and April 2015 carry errors of approximately 9.7%, compared to 2.6–3.4% for SARIMA in those months, because Prophet’s changepoint detection slightly overshoots at the training boundary before recovering.

The 95% confidence interval in Figure 11 is noticeably wide, reaching approximately $\pm\$1.5\text{M}$ by month 12, compared to SARIMA’s $\pm\$500\text{K}$. This is a direct consequence of multiplicative seasonality mode. In multiplicative mode, Prophet scales forecast uncertainty as a percentage of the trend level, so as the predicted demand baseline grows, the uncertainty band widens proportionally. This is the correct behaviour for a series where variance has grown alongside the mean, but it means that long-horizon planning intervals are wide and should be communicated to decision-makers as ranges, not point estimates.

8.4 Representative Series Results

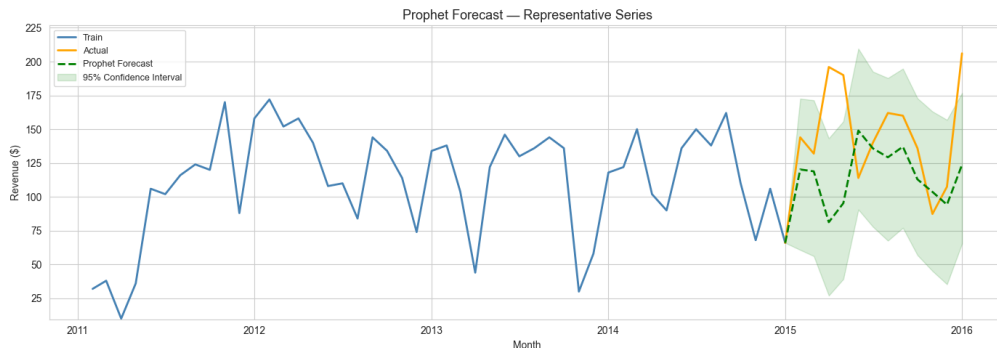


Figure 13: Prophet forecast ($\text{CPS} = 0.01$, additive seasonality) for the representative individual series. The stiff trend avoids overfitting to intermittent demand spikes. The same three months that produced the largest SARIMA errors (April 2015, May 2015, January 2016) produce comparable errors here, confirming a signal ceiling rather than model misspecification.

Table 7: Prophet representative series performance vs. all baselines (12-month test period).

Model	RMSE	MAE	MAPE
Naive	\$98.48	\$91.31	55.29%
SMA(3)	\$85.66	\$77.31	45.80%
SARIMA(0, 0, 1)(0, 1, 1) ₁₂	\$54.41	\$37.39	22.22%
Prophet (CPS=0.01, add.)	\$52.80	\$39.68	24.25%

Figure 13 and Table 7 show the individual series results. Prophet achieves a MAPE of 24.25% and an RMSE of \$52.80, against SARIMA’s 22.22% MAPE and \$54.41 RMSE. SARIMA wins on MAPE by 2 percentage points; Prophet wins on RMSE by \$1.61. Given the noise level of an individual product–store series (median 2 units per day, MAPE skewed by a handful of high-error months), this difference is not meaningful: two different model families have converged to essentially the same approximation accuracy.

More telling is the month-by-month error distribution. Prophet produces its three largest errors in April 2015 (58.5%), May 2015 (49.8%), and January 2016 (40.0%), the exact months where SARIMA failed most severely. This convergence across independent models is the key result: the approximation failures in those months are not caused by model misspecification in either framework. They reflect demand variation driven by external signals (price changes, SNAP distribution timing) that neither model has access to. This is formalised as the signal ceiling in Section 10.

9 Store-Level Analysis

9.1 Setup and Rationale

Three stores were selected for independent Prophet modeling: CA_1 (12.0% of platform revenue), CA_3 (17.1%), and TX_2 (10.9%). Each model was tuned via per-store cross-validation on training data only, the aggregate hyperparameters (CPS=0.5, multiplicative) were not assumed to transfer across hierarchy levels. All three stores share the 48-month train / 12-month test split used throughout.

Table 8 reports the CV-selected configurations. All three stores selected stiff trends ($\text{CPS} \leq 0.05$), the opposite of the aggregate winner. Individual store series carry more noise relative to signal than the 30,490-series aggregate; a flexible trend overfits to month-to-month variation rather than recovering the underlying trajectory. TX_2’s CV MAPE of 11.4%, measured entirely within the training window, signals that the model is already struggling to approximate this store’s demand from time series history alone.

Table 8: Prophet hyperparameters selected per store via cross-validation on training data.

Store	CPS	Mode	CV MAPE
CA_1	0.01	Multiplicative	4.9%
CA_3	0.01	Multiplicative	4.3%
TX_2	0.05	Additive	11.4%

9.2 CA_1: Smooth Demand, Strong Approximation

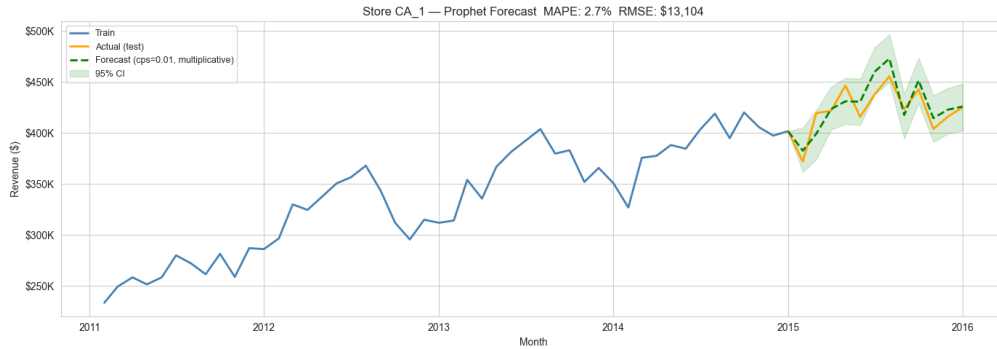


Figure 14: Prophet forecast for CA_1 (CPS=0.01, multiplicative) over the 12-month test period. Every month lands within 5.1% of observed demand; January 2016 error is 0.2%. CA_1’s smooth, consistent trajectory is the condition under which univariate baselines perform at their best.

CA_1 achieves 2.66% MAPE – the best result across any series at any hierarchy level in this project. The store’s observed demand through the test period is consistent with its training history, leaving no structural gap between what the model learned and what it was asked to project.

9.3 CA_3: Consistent Directional Bias

CA_3 achieves 6.23% MAPE with a consistent directional signature: the model overshoots in nearly every test month. The demand growth rate decelerated slightly in the test period, a regime shift no model calibrated on pre-2015 data could detect. A systematic overshoot is operationally more manageable than random error of the same magnitude, since planners can apply a known directional adjustment.

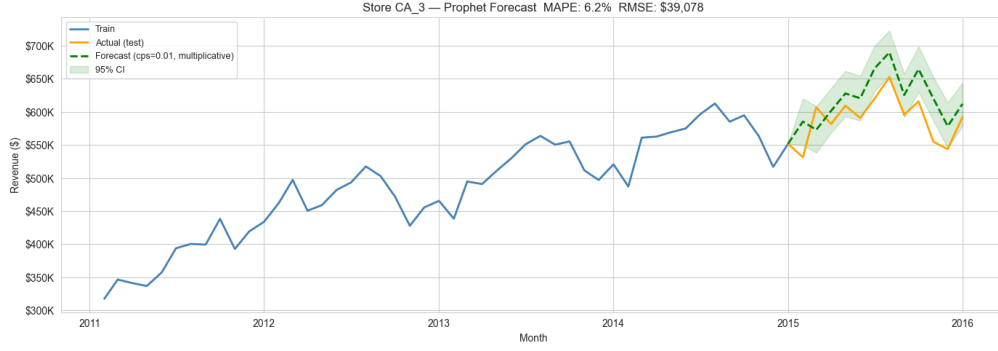


Figure 15: Prophet forecast for CA_3 (CPS=0.01, multiplicative) over the 12-month test period. The model overshoots observed demand in almost every month. CA_3’s demand growth decelerated slightly in the test period relative to training; November 2015 is the largest miss at 11.8%.

9.4 TX_2: The Clearest Store-Level Univariate Ceiling

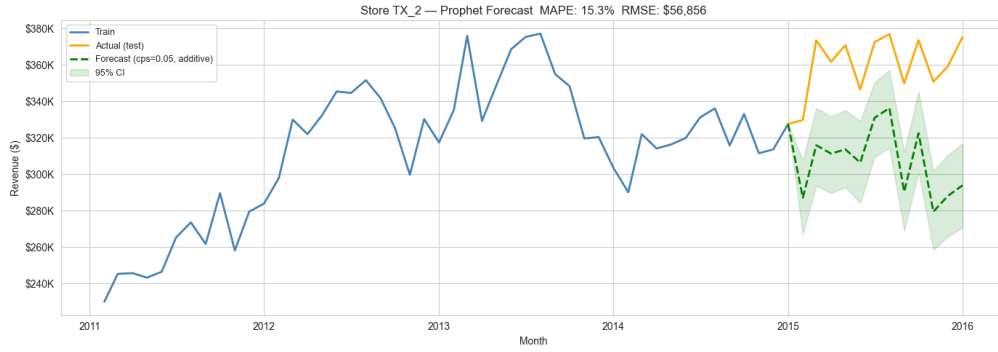


Figure 16: Prophet forecast for TX_2 (CPS=0.05, additive) over the 12-month test period. The model undershoots in every month, with the gap widening from 12.9% in February 2015 to 21.7% by January 2016; the signature of a systematic demand growth underestimate driven by unobserved external signals.

TX_2 achieves 15.31% MAPE, the weakest store-level result. Unlike the bounded bias on CA_3, TX_2 shows a progressively widening undershoot. The model falls further behind actual demand with each passing month. The EDA identified the most likely driver: TX FOODS demand shows a +17.2% uplift on SNAP distribution days, the strongest SNAP effect in the dataset. Prophet has no mechanism to receive the SNAP calendar as an explicit input. TX_2 is therefore the store where external feature inputs will produce the largest improvement over univariate baselines.

9.5 Cross-Store Variation

Table 9: Store-level Prophet results – 12-month test period.

Store	Revenue share	RMSE	MAE	MAPE
CA_1	12.0%	\$13,104	\$11,313	2.66%
CA_3	17.1%	\$39,078	\$36,472	6.23%
TX_2	10.9%	\$56,856	\$55,343	15.31%

The 12.65 percentage point spread across three stores in the same retail chain reflects genuine structural differences in demand composition, not noise. CA_1 and CA_3 have smooth California demand profiles; TX_2’s stronger SNAP-driven volatility exposes the limits of univariate modeling. These findings reinforce the EDA conclusion that state-level features are insufficient. Store-level structural differences dominate even within a single chain.

10 Signal Ceiling

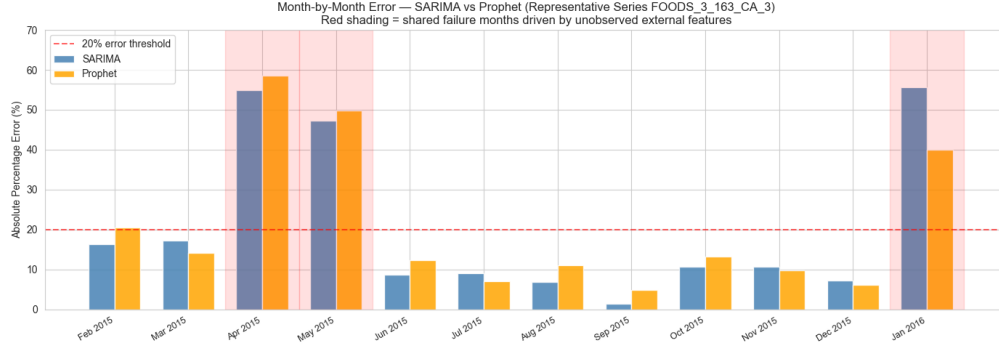


Figure 17: Month-by-month absolute percentage error for SARIMA and Prophet on the representative individual series. Red shading marks the three months where both models exceed 40% error: April 2015 (SARIMA 54.9%, Prophet 58.5%), May 2015 (SARIMA 47.3%, Prophet 49.8%), and January 2016 (SARIMA 55.6%, Prophet 40.0%). Two independent model families converging on the same failure months confirms these are external-driver events, not model misspecification.

In any forecasting system there exists an *irreducible error*: the portion of observed variation that no model with the current feature set can explain. For univariate time series models this includes all variation driven by inputs not present in the series: price changes, promotional events, and SNAP distribution timing.

Figure 17 makes this concrete. In 9 of 12 test months both models produce errors below 20% and broadly agree. The three shaded months are outliers in both series simultaneously. April and May 2015 are most consistent with a price-promotion event. The EDA confirmed that a 10% price drop at the product-store level is associated with approximately 77% more observed units ($r = 0.553$). January 2016 is most consistent with a SNAP calendar shift, California SNAP days produce a +10.3% FOODS revenue uplift, and a timing displacement across month boundaries is invisible to a model reading only past revenue values.

The signal ceiling on the representative series is approximately 22% MAPE. No parameter adjustment in either model closes this gap because the missing information does not exist in the revenue series either model reads. The path forward is encoding external signals directly as features, including product-store level price changes and state-level SNAP indicators.

11 Discussion

11.1 Full Results Summary

Table 10: All models – aggregate platform revenue (12-month test period).

Model	RMSE	MAE	MAPE
Naive	\$380,051	\$332,241	8.96%
SMA(3)	\$443,310	\$396,285	10.71%
SARIMA(2, 0, 1)(0, 1, 1) ₁₂	\$277,220	\$252,147	6.91%
Prophet (CPS=0.5, mult.)	\$209,726	\$178,567	5.02%

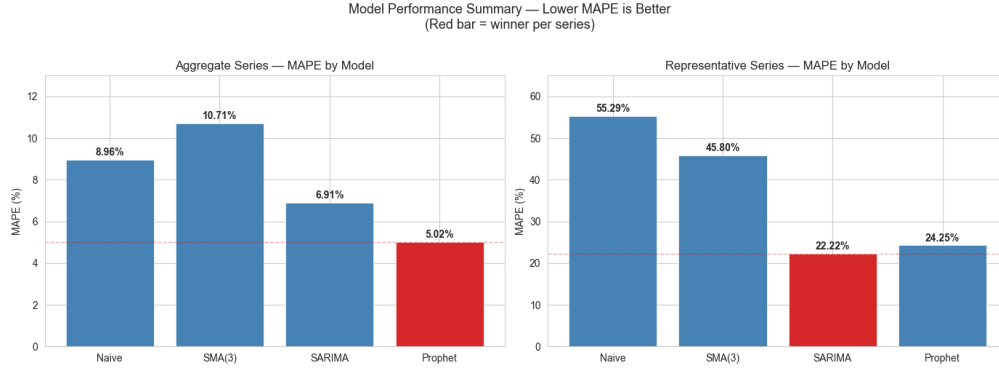


Figure 18: MAPE across all models and hierarchy levels for the 12-month held-out test period (Feb 2015 – Jan 2016). Lower is better. Aggregation determines the performance floor: the Naive baseline achieves 8.96% on the aggregate but 55.29% on the individual series. Prophet leads at the aggregate level; SARIMA and Prophet tie at the individual level; store-level results span 2.66% (CA_1) to 15.31% (TX_2).

Table 11: All models – representative individual series FOODS_3_163_CA_3 (12-month test period).

Model	RMSE	MAE	MAPE
Naive	\$98.48	\$91.31	55.29%
SMA(3)	\$85.66	\$77.31	45.80%
SARIMA(0, 0, 1)(0, 1, 1) ₁₂	\$54.41	\$37.39	22.22%
Prophet (CPS=0.01, add.)	\$52.80	\$39.68	24.25%

11.2 What the Results Reveal Collectively

Three findings emerge consistently across all hierarchy levels.

Aggregation determines the performance floor. The 46 percentage point gap between the Naive baseline on the aggregate (8.96%) and on the individual series (55.29%) is not a modeling result, it is a direct consequence of the law of large numbers. Summing 30,490 series cancels random demand shocks that no model can capture at the individual level. All four models benefit from this smoothing; the ordering among them reflects what each adds on top of it.

SARIMA and Prophet solve different problems. SARIMA’s advantage is short-horizon accuracy: errors of 1.7–3.4% in the first three test months on the aggregate reflect well-calibrated local structure. Prophet’s advantage is long-horizon accuracy: its piecewise linear trend holds to 1.3% error by month 12 where SARIMA degrades to 10.9%. The choice between them is a planning horizon decision, not a universal ranking. For planning cycles beyond six months, Prophet is the stronger tool on trending aggregate series.

The univariate ceiling is real and identifiable. At the individual series level, two independent model families converge at roughly 22% MAPE and fail on the same three months. This is not a coincidence, it is the measurable boundary of what historical revenue patterns can predict without external inputs. The same ceiling appears at the store level in TX_2, where the widening undershoot traces directly to unobserved SNAP signals. The ceiling is structural, not a tuning problem, and it points specifically to what additional features would close it.

12 Limitations

Several structural limitations constrain the conclusions that can be drawn from this analysis.

Observed sales are not true demand. Revenue is derived as $\text{units_sold} \times \text{sell_price}$ and used as a demand proxy throughout. This is a censored signal: when a product is out of stock, observed sales are zero regardless of

how many customers wanted it. Without inventory records or stockout logs, these two causes of zero observations are statistically indistinguishable. All reported metrics measure how well models approximate observed sales, not how well they recover latent demand.

Stockouts are unobservable. The zero-streak analysis showed that 44% of product-store series have a consecutive zero-sales run exceeding 365 days. These are almost certainly periods where the product was not stocked, not periods of genuine zero demand. Models trained on these series are learning partly from supply gaps rather than demand patterns, and their confidence intervals do not account for this source of uncertainty.

Intermittency at the individual level. At the product-store level, 68.2% of daily observations are zero and the non-zero distribution has skewness of 11.89. Monthly aggregation reduces but does not eliminate this problem. The Gaussian distributional assumption behind SARIMA’s likelihood estimation is strained at this level, making confidence intervals unreliable even when point estimates are reasonable.

Only 8.1% of series are modelable with SARIMA. SARIMA requires a continuous time index. The 2,469 complete series used in this project are not representative of the full 30,490, they are systematically the higher-volume, more stable products. Results for the representative series likely overstate what SARIMA could achieve on the average product-store combination.

The dataset ends in 2016. The training window covers February 2011 through January 2015; the test period covers February 2015 through January 2016. This predates significant changes in retail demand patterns, including the growth of e-commerce, shifts in consumer behaviour, and supply chain disruptions. Models calibrated on this window may not transfer to modern retail environments without retraining.

Univariate models cannot observe external demand drivers. As demonstrated by the signal ceiling in Section 10, price changes, SNAP distribution timing, and promotional events drive meaningful demand variation that no model reading only the revenue time series can capture. This is not a limitation of SARIMA or Prophet specifically; it is a fundamental constraint of the univariate modeling approach applied here.

13 Conclusion

This report addressed the following question: how well can classical time series models capture retail demand across aggregate, store, and individual SKU levels, and what structural limitations do they reveal?

The answer depends on the hierarchy level. At the platform aggregate, Prophet achieves 5.02% MAPE, a result strong enough to support budget planning and category-level inventory decisions. SARIMA achieves 6.91%, competitive at short horizons but degrading progressively beyond six months due to mean reversion. At the store level, results range from 2.66% MAPE on CA_1, where demand is smooth and consistent, to 15.31% on TX_2, where SNAP-driven volatility creates a structural gap that univariate models cannot close. At the individual product-store level, both SARIMA and Prophet converge near 22% MAPE, halving the naive baseline error but hitting a shared ceiling on the same three months.

The structural limitations revealed are equally important. Aggregation is not just a convenience, it is what makes classical forecasting viable. The law of large numbers transforms a sparse, intermittent, zero-inflated signal into a smooth, trend-dominated series that even simple heuristics approximate at under 10% MAPE. Disaggregating to the individual level removes this smoothing and exposes demand noise that no time series structure can explain.

The signal ceiling finding is the most precise conclusion of this analysis. When two independent model families fail identically on the same three months, the cause is not model misspecification, it is missing information. Price changes and SNAP distribution timing drove those demand events, and they exist in the dataset but outside the revenue

series that both models read. Classical time series models are not the wrong tool; they are the right tool applied to the wrong feature set. Extending them with external demand drivers, product-level price features and state-level SNAP calendar flags, is the logical and well-motivated next step.

Appendix

A Model Specifications and Hyperparameters

A.1 SARIMA Orders

Table 12: Final SARIMA specifications selected via AIC grid search on the 48-month training window.

Series	Selected order	Grid searched	Selection criterion
Aggregate	$(2, 0, 1)(0, 1, 1)_{12}$	$p \in \{0, 1, 2\}, q \in \{0, 1\}, P \in \{0, 1\}, Q \in \{0, 1\}$	AIC
Representative	$(0, 0, 1)(0, 1, 1)_{12}$	$p \in \{0, 1, 2\}, q \in \{0, 1\}, P \in \{0, 1\}, D \in \{0, 1\}, Q \in \{0, 1\}$	AIC

All models were estimated using `statsmodels.tsa.statespace.SARIMAX` with `enforce_stationarity=False` and `enforce_invertibility=False`. For the aggregate series, the two best AIC values differed by less than 0.01, indicating the selected specification is stable within the tested model space.

A.2 Prophet Hyperparameters

Table 13: Final Prophet configurations across aggregation levels.

Series	CPS	SPScale	HPScale	Mode
Aggregate	0.5	1.0	10.0	Multiplicative
Representative	0.01	1.0	10.0	Additive
CA_1	0.01	1.0	10.0	Multiplicative
CA_3	0.01	1.0	10.0	Multiplicative
TX_2	0.05	1.0	10.0	Additive

CPS = `changepoint_prior_scale`, SPScale = `seasonality_prior_scale`, HPScale = `holidays_prior_scale`.

Prophet models used yearly seasonality only (`weekly` and `daily` disabled), with a quarterly Fourier component added to capture medium-frequency variation. All models used `changepoint_range=0.8` and `interval_width=0.95`.

A.3 Holiday Event Windows

Table 14: Holiday windows used in Prophet models. Events not listed use a zero-width window.

Event	Lower	Upper
Christmas	-3	0
Thanksgiving	-3	0
NewYear	-1	0
SuperBowl	0	+1
LaborDay	0	+1
Easter	-1	+1
OrthodoxEaster	-1	+1

Window design follows observed demand timing patterns from the exploratory analysis, where holiday-related effects often appear slightly before or after the event day depending on whether the event restricts or accelerates consumption.

B Residual Diagnostics

B.1 Aggregate Series - SARIMA

Table 15: Residual diagnostics for $SARIMA(2, 0, 1)(0, 1, 1)_{12}$ on training data.

Test	p-value	Verdict
Ljung-Box (lag 10)	0.86	No autocorrelation
Jarque-Bera	0.44	Approx. normal residuals
Heteroskedasticity	0.88	Constant variance

Residual diagnostics indicate that remaining error behaves approximately like white noise, suggesting that the model has captured most linear temporal structure in the training window.

B.2 Representative Series - SARIMA

Table 16: Residual diagnostics for $SARIMA(0, 0, 1)(0, 1, 1)_{12}$.

Test	p-value	Verdict
Ljung-Box (lag 10)	0.84	No autocorrelation
Jarque-Bera	0.08	Mild skewness present
Heteroskedasticity	0.85	Constant variance

The representative series shows slightly non-normal residuals, consistent with intermittent demand behavior at the individual SKU level.

C Stationarity Test Results

Table 17: Augmented Dickey-Fuller tests across transformations.

Series	Transformation	ADF p-value	Result
Aggregate	Raw	0.318	Non-stationary
	First diff	0.059	Borderline
	Seasonal diff	< 0.001	Stationary
Representative	Raw	0.0004	Stationary
	First diff	< 0.001	Stationary
	Seasonal diff	< 0.001	Stationary

These results determine SARIMA differencing parameters used in the main model section.

D Month-by-Month Error Tables

The three highlighted months (April 2015, May 2015, January 2016) are identified as signal ceiling events: both model families produce errors exceeding 40% simultaneously. These months are the primary validation targets for the external-feature layer described in Section 10.

Table 18: Absolute percentage error by month: SARIMA and Prophet on the aggregate series (12-month test period, Feb 2015 – Jan 2016). Both forecasts are generated from models trained on Feb 2011 – Jan 2015 only.

Month	SARIMA APE (%)	Prophet APE (%)
Feb 2015	1.7	9.7
Mar 2015	3.4	9.7
Apr 2015	2.6	9.7
May 2015	5.5	6.5
Jun 2015	4.9	2.0
Jul 2015	6.1	1.2
Aug 2015	7.3	2.0
Sep 2015	7.9	2.1
Oct 2015	8.2	4.1
Nov 2015	9.1	3.8
Dec 2015	9.8	5.3
Jan 2016	10.9	1.3
MAPE	6.91	5.02

Table 19: Absolute percentage error by month: SARIMA and Prophet on the representative individual series FOODS_3_163_CA_3 (12-month test period). Shaded rows indicate months identified as signal ceiling events (both models exceed 40% error).

Month	SARIMA APE (%)	Prophet APE (%)
Feb 2015	16.3	20.5
Mar 2015	17.3	14.2
Apr 2015	54.9	58.5
May 2015	47.3	49.8
Jun 2015	8.7	12.3
Jul 2015	9.0	7.1
Aug 2015	6.9	11.0
Sep 2015	1.4	4.8
Oct 2015	10.7	13.2
Nov 2015	10.6	9.7
Dec 2015	7.2	6.1
Jan 2016	55.6	40.0
MAPE	22.22	24.25

E Data Preparation Notes

E.1 Aggregation Strategy

Revenue was constructed in a staged aggregation pipeline:

1. Aggregate units at store–item–week level.
2. Join weekly prices using store–item–week keys.
3. Compute revenue as units $\times$ price.
4. Aggregate to monthly and then to required hierarchy levels.

January 2011 is excluded due to partial coverage (3 days of data only).

E.2 Train/Test Split

- Training: Feb 2011 – Jan 2015 (48 months)
- Test: Feb 2015 – Jan 2016 (12 months)

No test data was used in model selection or tuning.

E.3 Representative Series Selection

The representative SKU (FOODS_3_163_CA_3) was selected from the set of complete series as the one closest to median total revenue across those series.

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